Jisoo Kim

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EDUCATION

PhD in Neuroscience, *University of Cambridge, UK* — Oct. 2021 - Present

Masters in Brain and Cognitive Engineering, *Korea University, Korea* — Sep. 2018 – Aug. 2020

Bachelor in Bio-system Medical Science, *Korea University, Korea* — Mar. 2014 – Aug. 2018

Exchange student, *University of Copenhagen, Denmark* — Dec. 2016 – July 2017

PUBLICATIONS

Jisoo Kim, Riccardo Beltramo, Jasper Poort
"Contributions of superior colliculus and primary visual cortex to visual spatial detection in freely moving mice."
Current Biology. (2026) https://doi.org/10.1016/j.cub.2026.01.077

Seoung-Jun Jo*, Hio-Been Han*, Dayoung Jung, **Jisoo Kim**, Jee Hyun Choi
"Mouse Escape Behaviors and mPFC-BLA Activity Dataset: Understanding Flexible Defensive Strategies Under Threat." ***Scientific Data.*** (2024) https://doi.org/10.1038/s41597-024-03688-0

Ana González-Rueda, Kristopher Jensen, Mohammadreza Noormandipour, Daniel de Malmazet, Jonathan Wilson, Ernesto Ciabatti, **Jisoo Kim**, Elena Wiliams, Jasper Poort, Guillaume Hennequin and Marco Tripodi
"Kinetic features dictate sensorimotor alignment in the superior colliculus."
Nature (2024) https://doi.org/10.1038/s41586-024-07619-2

Hio-Been Han, Hee-Sup Shin, Yong Jeong, **Jisoo Kim**, Jee Hyun Choi
"Dynamic switching of neural oscillations in the prefrontal–amygdala circuit for naturalistic freeze-or-flight."
Proc. Natl. Acad. Sci. (2023) https://doi.org/10.1073/pnas.2308762120

Jisoo Kim*, Chaewoo Kim, Hiobeen-Han, Jocheol-Jun, Wooseob Yeom, Sung Q Lee*, Jee Hyun Choi*,
"A bird's eye view of brain activity in socially interacting mice through mobile edge computing (MEC)"
Science Advances (2020) https://doi.org/10.1126/sciadv.abb9841

* Equally contributed as first authors

WORK EXPERIENCE

Maaind, *R&D intern, Neuro-adaptive AI startup* — June. 2024 – Aug. 2024

Decoded emotion from human EEG using unsupervised & supervised learning

Korea Institute of Science and Technology (KIST), *Graduate researcher* — Dec. 2017 – Aug. 2020

RESEARCH EXPERIENCE

University of Cambridge, *PhD student* — Oct. 2021 - Present

Behavioral Training: Visual Detection Task in Freely Moving Mice

- Developed a closed-loop system with DeepLabCut-live for controlling visual stimulus presentation.

Electrophysiology: Primary Visual Cortex (V1) and Superior Colliculus (SC)

- Performed simultaneous single-unit recordings in V1 and SC in freely moving mice.
- Classified cell types (putative PV, pyramidal neurons) using optogenetic-tagging method.

Optogenetics: Modulation of V1 and SC During Visual Detection

- Applied optogenetic inhibition to V1 and SC to assess their causal roles in visual detection.

Korea Institute of Science and Technology (KIST), *Graduate researcher* — Dec. 2017 – Aug. 2020

Effect of group condition on amygdala brain oscillations under robot-induced threat.

- Designed an innovative Robot-Induced Escape paradigm and discovered a new phenomenon.

Collaborative development of the wireless neural recording system.

- Validated the wireless recording system using rodent hippocampus theta oscillations.

TEACHING EXPERIENCE

University of Cambridge, *demonstrator*

IB (second year) Natural Sciences Tripos: Neurobiology: Function of Eye — Nov. 2024 – Dec. 2024

IB (second year) Natural Sciences Tripos: Neurobiology: Function of Eye — Nov. 2025 - Dec. 2025

FUNDING & AWARD

University of Cambridge, Wolfson college-PDN Studentship

Fully funded for four years — £25,000 per annum.

School of Biological Science Doctoral Training Program Award

Oversea student fee reduction award — £23,500 per annum.

Best Oral Presentation 2nd Prize

KIST Brain Science Institute (BSI) Symposium

SKILLS

Computational Skills

- MATLAB *(6+ years of experience)* and Python *(2+ years of experience)*
  Data analysis, noise removal, FFT, signal processing (Local Field Potential and single cell)
  power spectrogram, single cell sorting (Kilosort)
- **Machine Learning and Deep Learning** PyTorch, CNN, DeepLabCut
- **Software** Photoshop, Illustrator, Word, Excel, PowerPoint

Biological Skills

Stereotaxic rodent surgeries, Viral injection, 3D printing, Intraoperative optogenetic stimulation, Perfusion, Behavior experiments (Fear conditioning, Tube test, Robot-Induced Escape Experiment, Visual detection)

LEADERSHIP & VOLUNTEERING

Committee member *of the 2023 PDN Symposium, University of Cambridge* — Dec. 2021 – Apr. 2022

- Collaborated with researchers from diverse backgrounds to organize social programs.

Vice president *of Neuroscience society, New-Learn, Korea University* — Mar. 2015 – Dec. 2016

- Project team leader of the human psychology project and presented findings at a conference.

REFEREES

Jasper Poort

Asisstant Professor, Department of Physiology, Development and Neuroscience, University of Cambridge

Riccardo Beltramo

Asisstant Professor, Department of Physiology, Development and Neuroscience, University of Cambridge

JeeHyun Kwag

Professor, Department of Brain and Cognitive Sciences, Seoul National University

Jee Hyun Choi

Professor, Department of Bio-Medical Science and Technology, University of Science and Technology (UST)